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$$P_Y[M='hi'] = 0.3 \quad P_Y[M='no'] = 0.2 \quad P_Y[M='in'] = 0.5$$

$$1. \quad P_Y[M='hi' | C='st'] \\ = \frac{P_Y[C='st' | M='hi'] \cdot P_Y[M='hi']}{P_Y[C='st']}$$

$$\begin{cases} h \Rightarrow s: \text{shift } 11 \\ i \Rightarrow t: \text{shift } 11 \end{cases} \quad \begin{cases} n \Rightarrow s: \text{shift } 5 \\ o \Rightarrow t: \text{shift } 5 \end{cases} \quad \begin{cases} i \Rightarrow s: \text{shift } 10 \\ n \Rightarrow t: \text{shift } 6 \end{cases}$$

$$P_Y[C='st' | M='hi'] = P_Y[K=11] = \frac{1}{26}$$

$$P_Y[C='st' | M='no'] = P_Y[K=5] = \frac{1}{26}$$

$$P_Y[C='st' | M='in'] = 0$$

$$P_Y[C='st'] = 0.3\left(\frac{1}{26}\right) + 0.2\left(\frac{1}{26}\right) + 0.5(0) = \frac{1}{52}$$

$$\Rightarrow = \frac{\frac{1}{26} \cdot 0.3}{\frac{1}{52}} = 2 \cdot (0.3) = 0.6$$

$$2. \quad P_Y[M=b | C=4] \\ = \frac{P_Y[C=4 | M=b] \cdot P_Y[M=b]}{P_Y[C=4]}$$

$$= \frac{P_Y[K=k3] \cdot \frac{1}{4}}{\frac{1}{4} \cdot \frac{1}{4}}$$

$$= \frac{P_Y[C=4 | M=a] \cdot P_Y[M=a] + P_Y[C=4 | M=b] \cdot P_Y[M=b]}{\frac{1}{4} \cdot \frac{1}{4}}$$

$$= \frac{0\left(\frac{1}{4}\right) + P_Y[K=k3] \cdot \frac{1}{4}}{\frac{1}{4} \cdot \frac{1}{4}}$$

$$= \frac{\frac{1}{16}}{\frac{1}{4} \cdot \frac{1}{4}} = 1 \neq \frac{1}{4} = P_Y[M=b]$$

No, this encryption scheme is not perfectly secret because perfect secrecy requires $P_Y[M=m | C=c] = P_Y[M=m]$ for all m and c .

3. No. In the modified one-time pad scheme, if $k \neq 0^l$, $\Pr[M = m_i | C = m_i] = 0$ for a specific message m_i , because message and ciphertext are not allowed to be the same. However, $\Pr[M = m_i] > 0$ for that message m_i . Therefore, it breaks perfect secrecy requirement $\Pr[M = m | C = c] = \Pr[M = m]$ for all m and c .

4. Condition: $\Pr[\text{Enc}_k(m) = c] = \Pr[\text{Enc}_k(m') = c]$

$$\text{LHS} = \Pr[C = c | M = m]$$

Condition tells us that $\Pr[\text{Enc}_k(m) = c]$ is a fixed value for all message m . Therefore, $\Pr[C = c | M = m] = a$, where a is a constant value.

$$\begin{aligned} \Pr[C = c] &= \sum_{m \in M} \Pr[C = c | M = m] \cdot \Pr[M = m] \\ &= \sum_{m \in M} a \cdot \Pr[M = m] \\ &= a \sum_{m \in M} \Pr[M = m] \\ &= a \end{aligned}$$

Therefore, we have $\Pr[C = c | M = m] = \Pr[C = c]$

$$\begin{aligned} \Pr[M = m | C = c] &= \frac{\Pr[C = c | M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \quad \text{by Bayes' Rule} \\ &= \frac{1 \cdot \Pr[M = m]}{1} \\ &= \Pr[M = m] \end{aligned}$$

Therefore, it is perfectly secret.